

Purpose

This document defines how Butlerly validates usability, trust, accessibility, comprehension, and cross-device continuity before and after release. It establishes research methods, participant standards, test protocols, task scenarios, success measures, privacy protections, issue severity, decision rules, reporting, continuous discovery, and release gates for Version 1 across smartphone, tablet, and desktop.

1. Research Objective

Butlerly must be demonstrably understandable and usable by real people—not merely internally consistent. Research must determine whether users can complete essential tasks, understand what Butlerly and its AI are doing, maintain control of private data, recover from mistakes, and move between devices without losing trust.

Research must answer:

- Can users understand Butlerly's value and privacy model?
- Can they complete core tasks without training?
- Do they understand where data is stored and when it leaves a device?
- Can they distinguish source evidence, user-entered data, and AI inference?
- Can they correct, undo, recover, export, and delete confidently?
- Does the experience remain coherent across smartphone, tablet, and desktop?
- Can users with disabilities achieve equivalent outcomes?
- Which friction is acceptable protection, and which is unnecessary complexity?

2. Governing Principles

2.1 Test Outcomes, Not Screens

Research evaluates whether users accomplish meaningful goals. Screen-level observations are evidence, not the final unit of success.

2.2 Behavior Over Opinion

Observed behavior, errors, hesitation, recovery, and task completion carry more weight than preference statements alone.

2.3 Trust Is Measured

Trust is not assumed from visual polish. Research must test comprehension of privacy, AI uncertainty, data movement, device trust, synchronization, and reversibility.

2.4 Accessibility Is Core Research

Disabled participants and assistive-technology users are included throughout the product cycle, not only in a final audit.

2.5 Privacy by Research Design

Use fictional or participant-created test data. Never request real financial records, credentials, recovery secrets, or sensitive personal documents unless a separately approved study requires them.

2.6 Small Studies, Repeated Often

Frequent focused studies are preferred to rare, oversized studies. High-risk flows require deeper validation.

2.7 Findings Must Change Decisions

Research is complete only when findings have owners, priority, disposition, and traceable product changes or documented acceptance.

3. Research Program

Butlerly uses six complementary research layers.

Layer 1 — Concept Validation

Tests value proposition, terminology, privacy expectations, device strategy, and mental models before detailed interaction design.

Layer 2 — Structural Usability

Uses low-fidelity wireframes to test navigation, hierarchy, flow, state comprehension, and information scent.

Layer 3 — Interaction Usability

Uses clickable prototypes to test controls, sequence, errors, AI explanations, synchronization, and recovery.

Layer 4 — Accessibility and Inclusive Usability

Tests equivalent outcomes with screen readers, keyboard, switch access, voice control, large text, magnification, reduced motion, and other relevant access methods.

Layer 5 — Beta and Production Validation

Uses privacy-preserving product metrics, support themes, opt-in feedback, and targeted interviews to evaluate real-world performance.

Layer 6 — Longitudinal Trust Research

Evaluates whether Butlerly remains understandable and valuable after repeated use, corrections, synchronization events, device changes, and AI recommendations.

4. Participant Framework

Research must represent the intended Version 1 audience and meaningful edge cases.

Core participant dimensions:

- Personal-finance experience: novice, intermediate, advanced.

- Technology confidence: low, medium, high.
- Device pattern: smartphone-only, smartphone plus tablet, smartphone plus desktop, all three.
- Record-management behavior: manual, spreadsheet-based, app-based, document-heavy.
- Privacy sensitivity: moderate and high.
- AI familiarity: low, medium, high.
- Accessibility needs and assistive technologies.
- Age range, household context, language background, and cultural expectations relevant to the launch market.

Do not recruit only product experts, designers, engineers, or early adopters.

5. Sample Guidance

For focused formative tests, begin with 5–8 participants per materially different user segment or access method. Continue until major findings stabilize rather than treating a fixed number as proof of usability.

For comparative or quantitative studies, define sample size from the expected decision, baseline, effect size, confidence requirement, and risk. Do not present small-sample percentages as population truth.

Accessibility testing requires enough participants to cover materially different access strategies. One screen-reader user does not represent all blind users, and one keyboard user does not represent all motor-access needs.

6. Ethical and Privacy Requirements

Every study must document:

- Research purpose.
- Data collected.
- Recording method.
- Retention period.
- Access restrictions.
- Participant consent.
- Compensation.
- Withdrawal process.
- Handling of accidental sensitive disclosure.

Rules:

- Use fictional accounts and documents by default.
- Redact accidental private information immediately.
- Do not record passwords, biometric prompts, pairing secrets, recovery codes, or real financial identifiers.
- Store recordings and notes in approved restricted locations.
- Separate participant identity from research observations where practical.
- Delete raw research data according to the retention policy.

- Never use research participation to obtain broad product or marketing consent.

7. Study Brief Standard

Every study brief must include:

- Study ID and version.
- Decision to be informed.
- Research questions.
- Hypotheses.
- Target participants and exclusions.
- Method and session length.
- Prototype or build version.
- Platforms and devices.
- Accessibility configurations.
- Task scenarios.
- Success and failure criteria.
- Data captured.
- Privacy controls.
- Moderator guide.
- Analysis method.
- Owners and decision date.

Naming format:

UT-[year]-[sequence]-[topic]

Example: UT-2026-004-DEVICE-PAIRING

8. Moderation Standard

Moderators must:

- Use neutral language.
- Avoid teaching the interface before the task.
- Ask participants to explain expectations and interpretations.
- Distinguish spontaneous behavior from moderator-assisted behavior.
- Record critical incidents, hesitation, backtracking, errors, and recovery.
- Avoid asking “Do you like it?” as the primary question.
- Avoid leading questions such as “Was that easy?”
- Protect participants from embarrassment or blame.
- Pause if private information appears.
- Provide accessibility accommodations without changing the intended outcome.

9. Task Design Standard

Tasks use realistic goals, not interface instructions.

Preferred:

“You photographed a receipt yesterday and Butlerly is unsure about the category. Review it and save the correct record.”

Avoid:

“Open Review, select Needs Review, and accept the appropriate recommendation.”

Each task defines:

- Starting state.
- User motivation.
- Required outcome.
- Allowed paths.
- Critical errors.
- Assistance threshold.
- Completion criteria.
- Expected recovery paths.
- Privacy or accessibility considerations.

10. Core Version 1 Research Journeys

10.1 First Launch and First Value

Tasks:

- Explain what Butlerly does in the participant's own words.
- Explain where data is stored.
- Choose single-device or synchronized use.
- Complete the first useful record task.

Key measures:

- Value-proposition comprehension.
- Local-first comprehension.
- Time to first saved record.
- Consent accuracy.
- Abandonment reason.

10.2 Receipt Capture and Review

Tasks:

- Capture or import a receipt.
- Identify image-quality problems.
- Review extracted fields.
- Correct a low-confidence value.
- Save and locate the record.

Key measures:

- Capture completion.
- Field-correction accuracy.
- Evidence discoverability.
- Confidence comprehension.
- Recovery from processing failure.

10.3 Bulk Import

Tasks:

- Select a supported file.
- Review mapping.
- Resolve validation issues.
- Complete import.
- Find exceptions.

Key measures:

- Mapping accuracy.
- Error comprehension.
- Safe-resume success.
- Duplicate awareness.
- Import completion.

10.4 Review

Tasks:

- Identify what requires attention.
- Explain why an item matters.
- Review evidence.
- Accept, edit, defer, dismiss, or resolve appropriately.

Key measures:

- Correct prioritization.
- Decision accuracy.
- Evidence use.
- Unnecessary acceptance of AI suggestions.
- Review effort.

10.5 Search and Retrieval

Tasks:

- Find a record using remembered details.
- Apply filters.
- Inspect source evidence.
- Return to prior results without losing context.

Key measures:

- Search success.
- Query reformulation count.
- Filter comprehension.
- Result-source comprehension.
- Time to evidence.

10.6 Assistant and Explainability

Tasks:

- Ask a question about stored records.
- Identify the direct answer.
- Inspect supporting evidence.
- Describe uncertainty.
- Preview but do not accidentally execute an action.

Key measures:

- Answer comprehension.
- Evidence verification.
- Confidence calibration.
- Action-preview comprehension.
- False certainty or automation bias.

10.7 Add a Trusted Device

Tasks:

- Add a second device using an available method.
- Verify device identity.
- Understand what data will be copied or synchronized.
- Interrupt and resume initial synchronization.

Key measures:

- Pairing completion.
- Verification accuracy.
- Time to usable second device.
- Encryption and trust comprehension.
- Recovery from expiration or interruption.

10.8 Cross-Device Editing and Merge

Tasks:

- Edit different records on different devices.
- Edit different fields of the same record.
- Review a same-field resolution.
- Locate the earlier value in version history.

Key measures:

- Merge-model comprehension.
- Data-loss concern.
- Current-value identification.
- History discoverability.
- Confidence in continuity.

10.9 Offline and Provider Failure

Tasks:

- Capture and edit while offline.
- Identify what remains available.
- Understand pending changes.
- Recover after reconnection or provider degradation.

Key measures:

- Offline task completion.
- Saved-locally versus synchronized comprehension.
- Retry safety comprehension.
- Duplicate work avoided.

10.10 Privacy, Export, and Deletion

Tasks:

- Find where data is stored.
- Review external processing history.
- Export selected data.
- Delete and restore a record.
- Explain permanent deletion consequences.

Key measures:

- Data-location comprehension.
- Export-scope accuracy.
- Authentication completion.
- Recovery comprehension.
- Permanent-deletion error rate.

10.11 Lost Device and Revocation

Tasks:

- Identify a lost device.
- Revoke it.
- Explain what revocation does and does not do.
- Confirm remaining devices continue working.

Key measures:

- Correct device identification.
- Revocation completion.
- Remote-erasure misconception rate.
- Security confidence without false reassurance.

11. Accessibility Research Scenarios

At minimum, research must test:

- First launch with screen reader.
- Receipt review with large text and magnification.

- Full keyboard completion of import and record correction.
- Voice-control navigation of search and review.
- QR pairing alternative without camera dependence.
- Conflict comparison with screen reader.
- Device revocation with authentication fallback.
- Reduced-motion processing and synchronization states.
- High-contrast and non-color status comprehension.
- Error recovery when transient feedback is missed.

Accessibility outcomes must be reported by access method and task, not hidden inside one aggregate score.

12. Qualitative Measures

Capture:

- Mental model.
- Expected next step.
- Confidence before and after action.
- Perceived data location.
- Interpretation of AI role.
- Interpretation of synchronization state.
- Sources of hesitation.
- Recovery strategy.
- Trust gains and trust losses.
- Language that participants naturally use.
- Unmet needs and workarounds.

Research notes must distinguish observation, participant statement, researcher interpretation, and recommendation.

13. Quantitative Task Measures

For each core task, record where appropriate:

- Unassisted completion.
- Assisted completion.
- Failure.
- Time on task.
- Critical error count.
- Noncritical error count.
- Backtrack count.
- Assistance count.
- Abandonment.
- Recovery success.
- Confidence rating.
- Comprehension accuracy.
- Perceived effort.

Task completion definitions must be set before sessions begin.

14. Butlerly UX Metric Framework

14.1 Effectiveness

- Core task success rate.
- Correct-decision rate.
- Record accuracy after review.
- Search success rate.
- Pairing and revocation success.
- Recovery success.

14.2 Efficiency

- Time to first value.
- Time to capture and save.
- Time per review item.
- Time to retrieve source evidence.
- Time to usable second device.
- Steps and backtracks per task.

14.3 Comprehension

- Local storage comprehension.
- Saved-locally versus synchronized comprehension.
- AI evidence and confidence comprehension.
- Consent-scope comprehension.
- Merge and timestamp-resolution comprehension.
- Deletion and revocation consequence comprehension.

14.4 Trust and Control

- Ability to predict system behavior.
- Confidence in correction and recovery.
- Willingness to verify AI output.
- Perceived control over data movement.
- Trust after failure and recovery.
- False-confidence and automation-bias indicators.

14.5 Accessibility

- Task completion by access method.
- Assistive-technology critical errors.
- Keyboard-only completion.
- Large-text layout success.
- Accessible recovery success.
- Severity and age of accessibility defects.

14.6 Cross-Device Continuity

- Resume success.
- Merge success.
- Unresolved-conflict rate.
- Stale-data misunderstanding.
- Duplicate-work rate.
- Device-management comprehension.

15. Product Analytics Principles

Production analytics must be privacy-preserving and optional where required.

Permitted examples:

- Flow started, completed, canceled, failed, retried, or recovered.
- Time buckets rather than sensitive exact values.
- Screen or component IDs.
- Error categories.
- Pairing method.
- Offline or degraded state occurrence.
- Review-item type.
- Accessibility setting compatibility test results in controlled QA, not inferred disability status.

Prohibited analytics content:

- Record contents.
- Merchant names.
- Financial amounts.
- Document text or images.
- Assistant prompts or answers.
- Search terms.
- Encryption keys or pairing secrets.
- Device fingerprints beyond the minimum security requirement.
- Disability inference.

16. Metric Interpretation Rules

- Metrics do not replace observation.
- Faster is not always better when a task protects privacy or prevents data loss.
- High completion with low comprehension is not success.
- Repeated AI acceptance may indicate trust, convenience, or automation bias; investigate before interpreting.
- Aggregate metrics must not hide accessibility or platform failures.
- Compare like-for-like task definitions and product versions.
- Report uncertainty, sample size, and known bias.
- Do not optimize engagement time as a primary product goal.

17. Baselines and Targets

Before public release, establish baselines for every P0 journey. Initial targets must be ambitious but evidence-based after formative testing.

Non-negotiable release expectations:

- Zero known usability defects that create credible data-loss risk.
- Zero known accessibility blockers in core journeys.
- Users can identify whether work is saved locally, synchronized, or pending.
- Users can correct AI-proposed data without hidden acceptance.
- Users can complete export, deletion, revocation, and recovery with accurate consequence comprehension.
- Cross-device tests preserve valid changes.
- Trust-critical errors use understandable recovery language.

Numerical targets are maintained in the release scorecard and updated from evidence; they must not be invented solely to make a dashboard appear complete.

18. Issue Severity

Blocker

A core journey cannot be completed; a credible data-loss, privacy, security, or accessibility failure exists; or the interface causes a materially dangerous misunderstanding.

Critical

A core journey is possible only with major assistance, repeated failure, or misleading comprehension. Users may make consequential mistakes.

Major

A meaningful task has substantial friction, poor discoverability, or recoverable misunderstanding.

Minor

A localized issue creates limited friction without changing the outcome.

Observation

A pattern or opportunity worth tracking without current evidence of material harm.

19. Research Decision Rules

- Any Blocker requires resolution before release.
- Critical findings in P0 journeys require correction and retest, or documented executive exception with mitigation.
- Repeated major findings across participants require redesign or explicit scope reduction.
- A single serious privacy, security, accessibility, or data-integrity incident may outweigh frequency.
- Preference disagreement alone does not mandate change.
- Findings inconsistent with the design principles require root-cause analysis, not cosmetic adjustment.

- Retesting must use the revised experience, not only internal review.

20. Finding Format

Each finding includes:

- Finding ID.
- Concise title.
- Observed behavior.
- Participant evidence.
- Affected journeys, screens, and components.
- Frequency and participant segments.
- Severity.
- User impact.
- Trust, privacy, accessibility, or data-integrity impact.
- Root-cause hypothesis.
- Recommendation.
- Owner.
- Disposition: Fix, Investigate, Accept, Defer, or Out of Scope.
- Target release.
- Retest status.

21. Research Report Standard

Every report contains:

- Executive summary.
- Decision and research questions.
- Participants and limitations.
- Method and tested version.
- Task-level results.
- Findings by severity.
- Accessibility findings.
- Trust and privacy findings.
- Platform differences.
- Cross-device findings.
- Metric summary with uncertainty.
- Recommendations.
- Decision log.
- Issue register.
- Research artifacts and retention date.

Reports must not overgeneralize beyond the participants and method.

22. Continuous Discovery Cadence

During active product development:

- Conduct focused formative research at least once per significant interaction milestone.
- Test high-risk flows before engineering completion when prototypes are sufficient.

- Revalidate changed trust-critical components.
- Review support and feedback themes regularly during beta.
- Run accessibility regression studies at major release gates.
- Conduct longitudinal follow-up after users have experienced synchronization, corrections, and AI assistance over time.

Cadence may increase around onboarding, AI, privacy, synchronization, or major architecture changes.

23. Research Repository

Maintain a searchable repository containing:

- Study briefs.
- Moderator guides.
- Participant criteria.
- Consent records.
- Session notes.
- Findings.
- Clips or recordings where permitted.
- Decisions.
- Product changes.
- Retest outcomes.
- Metric definitions.

Every finding must link to the design or engineering work it informed. Sensitive participant information remains access-restricted and separated where practical.

24. Design and Engineering Integration

Before handoff:

- Research questions are attached to uncertain flows.
- Prototype hypotheses are explicit.
- Known risks are documented.

During implementation:

- Engineering flags behavior that differs from the tested prototype.
- Analytics events are reviewed for privacy.
- Accessibility behavior is verified on real devices.

Before release:

- P0 journeys are tested in the release candidate.
- Fixed Blocker and Critical findings are retested.
- Known Major issues have owners and plans.
- Metrics and support monitoring are ready.

After release:

- Production evidence is compared with research expectations.
- Unexpected behavior triggers focused study.
- Documentation is updated when evidence changes the standard.

25. Required Version 1 Study Sequence

Study 1 — Concept and Privacy Mental Model
 Study 2 — Smartphone Onboarding and First Capture
 Study 3 — Review, Evidence, and AI Correction
 Study 4 — Records, Search, and Source Retrieval
 Study 5 — Desktop Import and Exception Handling
 Study 6 — Tablet Review and Multi-Pane Comprehension
 Study 7 — Trusted Device Addition and Initial Sync
 Study 8 — Cross-Device Merge, Offline, and Recovery
 Study 9 — Privacy, Export, Deletion, and Revocation
 Study 10 — End-to-End Accessibility Validation
 Study 11 — Release Candidate P0 Journey Validation
 Study 12 — Beta Longitudinal Trust Study

The sequence may overlap, but no P0 area may reach release without appropriate validation.

26. Release UX Scorecard

The release scorecard includes:

- P0 journey completion status.
- Blocker and Critical findings.
- Accessibility defect status.
- Data-loss and integrity risks.
- Privacy and consent comprehension.
- AI calibration and correction results.
- Offline and synchronization results.
- Cross-device continuity results.
- Recovery results.
- Platform-specific failures.
- Open Major findings and owners.
- Retest evidence.
- Go, Conditional Go, or No-Go recommendation.

A single composite UX score must not conceal critical failures.

27. Roles and Governance

Design owns research quality, interaction findings, and design response.

Product owns prioritization, scope, and outcome decisions.

Engineering owns implementation fidelity and technical instrumentation.

QA owns independent verification and regression coverage.

Privacy and Security review trust-critical studies and telemetry.

Accessibility specialists and disabled participants validate inclusive outcomes.

Research independence must be protected. Teams may challenge findings, but they may not suppress evidence because it conflicts with schedule or preference.

28. Dependencies

UX-0012 depends on:

- UX-0001 — Butlerly Experience Blueprint.
- UX-0002 — Butlerly Version 1 User Journey Map.
- UX-0003 — Butlerly Information Architecture and Navigation Model.
- UX-0004 — Butlerly Core User Flows.
- UX-0005 — Butlerly Screen and State Inventory.
- UX-0006 — Butlerly AI Interaction and Explainability Specification.
- UX-0007 — Butlerly Design System Foundations.
- UX-0008 — Butlerly Cross-Device Experience and Continuity.
- UX-0009 — Butlerly Accessibility Standard.
- UX-0010 — Butlerly Low-Fidelity Wireframe Specification.
- UX-0011 — Butlerly Design Patterns and Component Library.

UX-0012 governs:

- Prototype research plans.
- Release-candidate usability validation.
- Production UX metrics and dashboards.
- Design QA priorities.
- UX issue severity and release recommendations.
- Continuous discovery and longitudinal research.

29. Acceptance Criteria

This specification is complete when:

- Every P0 journey has defined research tasks and measures.
- Trust, privacy, AI, synchronization, recovery, and accessibility are directly tested.
- Participant recruitment covers meaningful device, skill, privacy, and accessibility differences.
- Study briefs, moderation, task design, finding, and report standards are explicit.
- Product analytics rules prohibit sensitive content collection.
- Metric interpretation avoids engagement-only and speed-only optimization.
- Issue severity and release decision rules are enforceable.
- Findings connect to owners, product changes, and retesting.
- A complete Version 1 study sequence and release scorecard are defined.
- Product, Design, Engineering, QA, Privacy, Security, and Accessibility teams can use the document without inventing research policy.

All prototypes and builds used for usability testing must conform to BDS-0000 — Butler Design System v1.0 for visual tokens, theme values, typography, spacing/radius, responsive breakpoints, component/state behavior, accessibility implementation, localization/original-data handling, and visible navigation. Finance V1 primary destinations are Home, Transactions, Review, Search, and Settings; Assistant is optional/non-primary.

Finance V1 release-gate research prioritizes complete single-device local-first workflows. Studies covering trusted-device pairing, cross-device synchronization, merge/conflict behavior, revocation, or cross-device continuation remain useful future/optional research scenarios but are not MVP release blockers unless product scope explicitly enables those capabilities. No research scenario should imply that a Butlerly account is required for core operation.

Localization research must verify that UI-authored content switches with the selected language while user-entered/imported/scanned/source data retains its original language/script, and that original financial currency/amount remains distinguishable from any separately displayed reference conversion.